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Supplemental information

High-resolution dynamic imaging of chromatin DNA communication using Oligo-LiveFISH

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Table S1. Details of templates, primers, Oligo-LiveFISH probes, and FISH probes, related to Figures 1-7 and STAR Methods.

A common template used for generating crRNA or sgRNA is the following:

Adapter +**T7 promoter** + crRNA/sgRNA spacer+ crRNA/sgRNA common region

(1) crRNA oligo pool from Twist Bioscience (N is any nucleotide and varies from 12-19 for oligo pools; the spacer is underlined)

Template: Adapter + **TAATACGACTCACTATA**+GNNNNNNNNNN+GTTTAAGAGCTATGCTGTTTTG

Forward primer: Adapter sequence

Reverse primer: CAAAACAGCATAGCTCTTAAAC

(2) crRNA oligo pool from IDT (N is any nucleotide and varies from 12-19 for oligo pools; the spacer is underlined)

Forward primer: **TAATACGACTCACTATA**+GNNNNNNNNNN+GTTTAAGAGCTATGCTG

Reverse primer: CAAAACAGCATAGCTCTTAAAC

(3) sgRNA oligo pool from Twist Bioscience (N is any nucleotide and varies from 12-19 for oligo pools; the spacer is underlined):

Template:
Adapter+**TAATACGACTCACTATA**+GNNNNNNNNNNNN+GTTTAAGAGCTATGCTGGAAACAGCATAG
CAAGTTTAAATAAGGCTAGTCCGTTATCAACTTGAAAAAGTGGCACCGAGTCGGTGCTTTTTTT

Forward primer: Adapter sequence

Reverse primer: AAAAAAAGCACCGACTCGGTGCCAC

(4) DNA template for sgRNA for repetitive region (N is any nucleotide; the spacer is underlined):

Template:
GAAAT+**TAATACGACTCACTATA**+NNNNNNNNNNNN+GTTTAAGAGCTATGCTGGAAACAGCATAGC
AAGTTTAAATAAGGCTAGTCCGTTATCAACTTGAAAAAGTGGCACCGAGTCGGTGCTTTTTTT

Forward primer: GAAAT+**TAATACGACTCACTATA**+NNNNNNNNNNNN+GTTTAAGAGCTATGCTG

Reverse primer: AAAAAAAGCACCGACTCGGTGCCAC

(5) Adapter Sequences[1]:

Adapter 1: CTTGGTCAGACGAGTGC

Adapter 2: TGGTACGGGAACAGCAC

Adapter 3: CTCACCGCTCTTGTAGC

Adapter 4: AAGACGGCTGAGCCATC

(6) Oligo-LiveFISH crRNA, tracrRNA and sgRNA sequences:

dCas9 tracrRNAs:

5’- GGAACCAUUCAAAACAGCAUAGCAAGUUUAAAUAAGGCUAGUCCGUUAUC

AACUUGAAAAAGUGGCACCGAGUCGGUGCUUUUUUU-3’

dCas9 crRNAs (N is any nucleotide and varies from 13-20 for oligo pools; the spacer is underlined):

5’-NNNNNNNNNNNNGUUUAAGAGCUAUGCUGUUUUG-3’

dCas9 sgRNAs (N is any nucleotide and varies from 13-20 for oligo pools; the spacer is underlined):

5’-NNNNNNNNNNNNGUUUAAGAGCUAUGCUGGAAACAGCAUAGCAAGUUUAAAUAAGGC

UAGUCCGUUAUCAACUUGAAAAAGUGGCACCGAGUCGGUGCUUUUUUU-3’

(7) Oligo-LiveFISH and FISH probes used in the study:

crRNA/sgRNA	**Spacer Seq**	**repeats**	**Oligo-LiveFISH probe position (hg38)**	**FISH probe position(hg38)**	**References**
ChrX-*rep*	TAGAGTCTGCAT	x666	ChrX:26096-44821	ChrX:10001-44814 (WI2-1786K17)	This study
ChrX-*PLCXD1-nonrep*	Table S2	non-repetitive	ChrX:275569-283797 (96 probes, IDT)	ChrX:10001-44814 (WI2-1786K17)	This study
Chr13-*rep*	ACCATTCCTTC	x358	Chr13:112277011 - 112319170	Chr13:112276804-112319743 (WI2-438L18)	Ma et al., 2016[2]
Chr13-*TUBGCP3*	Table S2	non-repetitive	Chr13:112577342-112588167 (96 probes, IDT)	Chr13:112276804-112319743 (WI2-438L18)	This study
Chr3-*ACAP2-nonrep* (site1 195.5Mb)	Table S2	non-repetitive	Chr3:195446441-195470879 (192 probes, IDT)	Chr3:195434695-195587044 (RP11-165B7)	This study
Chr3-*ACAP2-nonrep* -site1 195.5Mb	Table S2	non-repetitive	Chr3: 195458383-195478290 (136 probes, Twist Bioscience)	Chr3:195434695-195587044 (RP11-165B7)	This study
Chr3-*ACAP2-rep* -site2 195.5Mb	TGATATCACAG	x529	Chr3:195478317 - 195506985	Chr3:195434695-195587044 (RP11-165B7)	Ma et al., 2016[2]
Chr3-site3 195.7Mb	AGGTATGGGTG	x116	Chr3:195775389-195777939	N.A.	Gao et al., 2021[3]
Chr3-site4 198.1Mb	TGGACAGTGGGA	x133	Chr3:198098870-198105828	N.A.	This study
MUC4 gene	Table S2	non-repetitive	Chr3: 195808706-195815721 (96 probes, IDT)	N.A.	This study
FOS gene promoter	Table S2	non-repetitive	Chr14:75238783-75278649 (237	Chr14:75161753-75339259 (787H13)	This study

			probes, Twist Bioscience)		
FOS gene enhancer	Table S2	non-repetitive	Chr14:75044178-75084248 (343 probes, Twist Bioscience)	Chr14:75161753-75339259 (787H13)	This study
MYC gene promoter	Table S2	non-repetitive	Chr8:127696155-127736034 (340 probes, Twist Bioscience)	N.A.	This study
MYC gene enhancer	Table S2	non-repetitive	Chr8:127650882-127691147 (375 probes, Twist Bioscience)	N.A.	This study
Telomere	GUUAGGGUUAGGGUUAGGGUUA	N.A.	N.A.	N.A.	Chen et al.,2013[4]
Non-targeting	GAACGACTAGTTAGGCGTGTA	N.A.	N.A.	N.A.	

Table S3. Comparison of microscopes used in the study, related to Figures 1-7 and STAR Methods.

Microscope	A1	A2	A3	A4	B	C
Modality	*Widefield Microscope*				*Spinning Disk Confocal Microscope*	*Double-helix Point Spread Function (DHPSF) Microscope*
Dimensionality	2D				2D and 3D	3D with double-helix phase masks
Commercialization	commercial	custom built			commercial	custom built
Number of colors	Up to 4 sequentially	1	2 simultaneously		Up to 4 sequentially	2 simultaneously
Advantages	High spatiotemporal resolution in 2D;	High spatiotemporal resolution in 2D; single-molecule sensitivity			High resolution in 2D; extract z position from z slices; pinholes reduce out-of-focus background;	High spatiotemporal resolution in 3D; No need to scan for z axis position over 2 μm axial range
Limitations	No axial position information; out-of-focus background				3D imaging has low time resolution due to dark periods for filter changes (time-lapse), and low z resolution	Up to two-color currently; out-of-focus background
Data in the paper	Fig.1J, 3D-E, 4E-H, S2C, S5B	Fig.5 B,5E, S6A-C S2A-B,	Fig.3 O-P, S5M	Fig.5F-J, 6I-J,5O S6D-F, S6M-N, S7H	Fig.1B-H, 2B-H, 3B, 3F, 3H-N, 4B-D, 4H, 4J, 5D, 6B-H, 6K, 7D-E, S1A-G, S2D-G, S3, S4, S5A, S5F-I, S5N, S7A-G, S7I-L	Fig.5L-O, 7F-P, S6G-L, S6O, S7M-N

Table S4. The communication time and effective genomic communication distance calculated from imaging and sequencing methods, related to Figures 1-7 and STAR Methods.

	Real genomic distance *s* (kb)	Communication time $\tau_R(\Delta n)$ from Oligo-LiveFISH using VCC (s)		Predicted communication time $\tau_R(\Delta n)$ via 1D-cis communication from Eq16 (s)	Contact probability *P(s)* from Hi-C	Genomic communication distance *s* from Oligo-LiveFISH (kb)	Genomic communication distance *s* from Hi-C (kb)
		2D VCC	3D VCC				
Sites1&2	28	0.44	---	---	0.073	28	28
Sites 2&3 canonical group	284	89	113	76	0.010	305	357
Sites 3&4 canonical group	2327	260	383	8108	0.0089	495	406
Sites 2&3 variant group	284	384	687	76	0.010	590	357
Sites 3&4 variant group	2327	238	362	8108	0.0089	475	406

Reference

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